

JC2001 INTRODUCTION TO SOFTWARE ENGINEERING (2026–27)

BSc in Business Management and Information Systems (BSc BMIS) — Group 5

PROJECT PROPOSAL

To Design and Implement a Smart Study Assistant System for
University Students

A GraphRAG and Knowledge Graph Diagnostic Approach to Misconception Remediation and Exam Preparation

Academic Supervisor: Dr. Shahzad Mumtaz (shahzad.mumtaz@abdn.ac.uk) | Submission Date: 25 September 2026

Team Member Roster (Group 5)

No.	Full Name	Student ID	Designated Project Role	University Email
1	Yuxuan Wu	50106070	Project Manager & Team Lead	u11yw25@abdn.ac.uk
2	Yongtong Lin	50106038	Principal Concept Co-Originator & Lead Domain Analyst	u19yl25@abdn.ac.uk
3	Sijian Wang	50106045	System Architect (UML & Knowledge Graph)	u14sw25@abdn.ac.uk
4	Hao Jiang	50106065	PoC Lead Developer (GraphRAG & Neo4j Engine)	h.jiang.25@abdn.ac.uk
5	Weixin Xie	50106034	Principal Concept Co-Originator & QA Lead	u02wx25@abdn.ac.uk
6	Zijian Zhang	50106035	Business Analyst (User Stories & Acceptance)	u17zz25@abdn.ac.uk
7	Yusai Xi	50105989	UI/UX Designer & Wireframe Lead	u02yx25@abdn.ac.uk
8	Siqin Dong	50106060	PoC Logic Developer (Diagnostic API & Quiz)	u20sd25@abdn.ac.uk
9	Mingjie Yang	50106061	Software Testing & User Manual Lead	u05my25@abdn.ac.uk
10	Zixuan Liang	50106037	Presentation, Media & Deployment Lead	u14zl25@abdn.ac.uk

1. Project Title

To Design and Implement a Smart Study Assistant System for University Students

A GraphRAG and Knowledge Graph Diagnostic Approach to Misconception Remediation and Exam Preparation

This project aims to help university students deal with revision fatigue and the challenge of mastering large, complex syllabi. Most existing tools focus on passive tasks—storing documents or running keyword searches—but they do little to help students actually understand where their knowledge breaks down. Our approach is different. We combine Retrieval-Augmented Generation (GraphRAG) [1], [2] with structured knowledge graphs built in Neo4j [3] to represent courses as networks of concepts, prerequisites, and common misconceptions. Drawing on Bloom's two-sigma tutoring research [4] and Sweller's Cognitive Load Theory [5], the system is designed to move students beyond rote memorization toward a deeper, more self-aware form of exam preparation.

2. Problem

Undergraduate students face several recurring problems during exam preparation, and we identified four that are especially damaging.

The most immediate is **cognitive overload from fragmented materials** [5]. Syllabi spread concepts across lecture slides, textbook chapters, and tutorial sheets with no unifying structure. Students end up spending more time locating information than actually learning it.

A related issue is **superficial memorization**. Conventional flashcards encourage students to recall isolated definitions without understanding how ideas connect. This creates a false sense of confidence—what psychologists call the illusion of explanatory depth—and leaves students unprepared for questions that require multi-step reasoning [6].

There is also a lack of **meaningful diagnostic feedback** [7]. When students practise past exams, they usually get a score and a set of correct answers. That tells them what they got wrong, but not why. It does not explain which misconception led to the error or which prerequisite concept needs revisiting.

Finally, general-purpose Large Language Models (LLMs) are unreliable in academic settings due to **knowledge hallucination** [2]. In our own preliminary testing, unguided LLMs produced modest accuracy in analytical subjects (61.00% in Law, 64.91% in Econometrics, 60.26% in Physics), and frequently generated fabricated prerequisites, flawed derivations, or

made-up counterexamples. Without structured knowledge constraints, these models can actively mislead students rather than help them.

3. Solution

Group 5 proposes the **Smart Study Assistant System**, coupling GraphRAG retrieval with Neo4j cloud graph topology, automated diagnostic parsing, Socratic tutoring, dynamic quiz generation, and interactive radar visualization. Crucially, the system directly reduces academic knowledge hallucination rates by grounding generative reasoning onto verified conceptual and prerequisite subgraphs.

3.1 Neo4j Knowledge Graph Schema with Misconception Modeling

The system builds on a property graph in Neo4j [3]. Nodes labeled **:Concept** span four functional categories in English: **Core Concept**, **Prerequisite Condition**, **Misconception / Trap**, and **Operational Step**. The primary architectural innovation is the **:DEPENDS_ON** relationship (with property **type: "PRONE_TO_MISCONCEPTION"**), alongside **PREREQUISITE_OF**, **LOGICAL_DERIVATION**, and **COMPUTATIONAL_STEP**. Cypher upsert logic (**MERGE ... ON MATCH SET**) cumulatively aggregates distinct examination traps into enriched conceptual nodes without entity duplication.

3.2 Automated Diagnostic Ingestion via Qwen-Turbo

An automated pipeline using the DashScope Qwen-Turbo API ingests multi-disciplinary exam questions. Acting as a logic diagnostic engine rather than a generic answer generator, the model parses multiple-choice and analytical problems into structured JavaScript Object Notation (JSON) format specifying core concepts, strict boundary conditions, and distractor fallacy explanations. Exponential backoff with jitter mitigates HTTP 429 rate limits, and regex sanitization ensures robust JSON extraction.

3.3 GraphRAG Socratic Tutoring Engine

To eliminate passive answer copying, the backend implements an active Socratic tutoring engine. When querying a topic, the GraphRAG pipeline retrieves deterministic subgraphs from Neo4j (**LIMIT 40**) constructing explicit logic chains (e.g., (Concept A) -> PRONE_TO_MISCONCEPTION -> (Trap B)). The tutor is strictly constrained to withhold direct answers, instead using graph facts to pose focused Socratic questions (under 150 words) compelling students to identify missing boundary conditions and self-correct [4], [6].

3.4 Dynamic Quiz Generation & ECharts 5.5.1 Mastery Visualization

The adaptive quiz engine traverses the knowledge graph to detect fragile prerequisite nodes, dynamically assembling diagnostic quizzes whose distractors map to known misconceptions. Choosing an incorrect option triggers immediate, targeted remediation. The Single Page Application (SPA) frontend integrates Apache ECharts 5.5.1 to project a five-dimensional student mastery radar chart (Basic Recall, Boundary Deduction, Trap Defense, Cross-Topic Synthesis, Calculation Precision) with dark (**#1e293b**) / light (**#ffffff**) theme switching.

3.5 Empirical Benchmark Grounding & Noise Arbitration Gate

The proposed architecture is informed by preliminary proof-of-concept benchmarks across twelve academic corpora conducted during the proposal phase. Grounding retrieval in explicit Neo4j knowledge graphs directly suppresses hallucination rates and yields immediate accuracy gains in structured disciplines: **+2.50% in Professional Law** (61.00% to 63.50%), **+2.00% in High School Computer Science** (86.00% to 88.00%), **+1.82% in Public Relations** (72.73% to 74.55%), and **+1.75% in Econometrics** (64.91% to 66.67%). To eliminate knowledge noise and combat academic hallucinations in narrative domains (e.g., Physics divergence), the planned system enforces an Academic Arbitration Gate (**RAG_CONFIDENCE_THRESHOLD = 85**), pruning spurious nodes and hallucinated causal leaps before prompt synthesis.

4. Objectives

Group 5 defines five concrete objectives adhering to the SMART framework (Specific, Measurable, Achievable, Relevant, Time-bound) concluding on or before **14 December 2026**:

- 1. Objective 1 (Functional PoC Delivery):** Design, develop, test, and containerize a functional Proof-of-Concept Smart Study Assistant system by **14 December 2026**, integrating Neo4j cloud storage, Qwen-Turbo diagnostics, Socratic tutoring, and ECharts 5.5.1 visualization into a runnable bundle.
- 2. Objective 2 (Knowledge Graph Scale & Latency):** Construct a domain graph comprising at least 250 concept nodes, 500 exam questions, and 300 explicit misconception relationships (**PRONE_TO_MISCONCEPTION**), maintaining sub-2.0-second (<2000ms) graph traversal latency on bidirectional Cypher queries (depth ≤ 2 , **LIMIT 40**).
- 3. Objective 3 (Diagnostic Precision & Hallucination Rate Reduction):** Achieve automated diagnostic extraction accuracy of $\geq 85\%$ on exam distractors, enforce a strict confidence arbitration threshold (≥ 85) filtering out at least 85% of spurious graph

relationships, and substantially curtail AI hallucination rates in academic tutoring through deterministic knowledge grounding.

4. Objective 4 (Software Quality & Test Coverage): Engineer the platform under modular object-oriented principles [8], achieving 100% unit test coverage on core Cypher query wrappers, $\geq 80\%$ branch coverage across diagnostic logic with pytest, zero unhandled HTTP 429 exceptions, and full PEP 8 compliance.

5. Objective 5 (Documentation Compliance): Deliver all mandatory JC2001 assessment deliverables by **14 December 2026 at 23:59 CST**, delivering a 40–60 page Technical Report (without raw code), an 8–12 page User Manual PDF, a 15-minute MP4 presentation video with slides, and an Appendix A Workload Profile signed by all ten members.

5. Benefits

The system delivers measurable value across key educational stakeholders:

5.1 Benefits to Students (Primary Stakeholders)

- **Reduced Revision Overhead:** Automated structuring of fragmented notes into a coherent concept graph reduces study collation time by 30–40%.
- **Targeted Trap Remediation:** Mapping questions to directional misconception edges (**PRONE_TO_MISCONCEPTION**) reveals root causes of error, preventing repetitive mistakes [6].
- **Metacognitive Retention:** Socratic dialogue compels students to articulate boundary conditions rather than passively memorizing answers [4].
- **Visual Competency Tracking:** Five-dimensional ECharts radar analytics provide transparent feedback on knowledge gaps, mitigating exam anxiety.

5.2 Benefits to Instructors & Teaching Assistants

- **Cohort Misconception Heatmaps:** Aggregated diagnostic logs expose systemic curriculum blind spots, guiding targeted tutorial review.
- **Automated Quiz Generation:** Instructors can dynamically generate diagnostic quizzes aligned with syllabus learning outcomes.
- **Consultation Efficiency & Institutional Value:** Automated remediation of recurring misconceptions frees teaching assistants for higher-order academic mentoring while establishing reusable digital curriculum assets.

6. Timeline

The project schedule spans four chronological phases from 14 September 2026 to 14 December 2026: Phase 1 (Requirements & Proposal), Phase 2 (Architecture & UML), Phase

3 (PoC Implementation & Verification), and Phase 4 (Final Documentation & Presentation), strictly aligned with JC2001 course milestones:

Project Milestones Schedule

Phase	Milestone ID	Milestone Description & Deliverables	Target Completion Date
Phase 1	M1.1	Group Formation, Leadership Election & MyAberdeen Enrolment	14 September 2026
	M1.2	Supervisor Allocation & Formal Kick-off Meeting Briefing	18 September 2026
	M1.3	Formal Project Proposal Submission on MyAberdeen (PDF)	25 September 2026
Phase 2	M2.1	Requirements Traceability Matrix & Formal Use Case Modeling	10 October 2026
	M2.2	UML Architectural Design Package & Neo4j Ontology Baseline	20 October 2026
	M2.3	Mandatory Project Update 1 Submission on MyAberdeen	23 October 2026
	M2.4	Mid-Term Architecture Baseline & Supervisor Check-in	31 October 2026
Phase 3	M3.1	Mandatory Project Update 2 Submission on MyAberdeen	06 November 2026
	M3.2	GraphRAG Engine & Cloud Neo4j AuraDB Integration	15 November 2026
	M3.3	Dynamic Quiz Generation & ECharts 5.5.1 Mastery Dashboard	22 November 2026
	M3.4	Mandatory Project Update 3 Submission on MyAberdeen	25 November 2026
	M3.5	Multi-Discipline Empirical Benchmarking & Verification Suite	30 November 2026
Phase 4	M4.1	Optional Project Update 4 Buffer Submission on MyAberdeen	04 December 2026
	M4.2	Technical Report Master Draft (40–60 pages) & Workload Review	08 December 2026
	M4.3	User Manual PDF (8–12 pages) & PoC Runnable Bundle Packaging	10 December 2026
	M4.4	15-Minute MP4 Video Recording & Final Presentation Slides	12 December 2026
	M4.5	Final Course Deliverables Submission on MyAberdeen	14 December 2026

7. Action Plan

The foundational research concept and diagnostic problem formulation of this project were originally initiated by **Yongtong Lin** and **Weixin Xie** (*Principal Concept Co-Originators*).

To ensure balanced workload distribution across all ten group members throughout the 12-week development lifecycle, the Work Breakdown Structure (WBS) assigns specific engineering activities linked directly to project objectives:

Work Breakdown Structure (WBS) Matrix

WBS ID	Action Activity & Technical Deliverable	Linked Objective	Responsible Member(s)	Student ID	Deadline	Progress Status
WBS 1.1	Requirements Elicitation & Domain Concept Modeling	Obj 1, 5	Yongtong Lin, Zijian Zhang	50106038, 50106035	2026-09-25	Completed
WBS 1.2	Project Proposal Governance, Editing & Formatting	Obj 5	Yuxuan Wu, Weixin Xie	50106070, 50106034	2026-09-25	Completed
WBS 2.1	Neo4j Graph Ontology & PRONE_TO_MISCONCEPTION Schema Specification	Obj 2	Sijian Wang, Hao Jiang	50106045, 50106065	2026-10-20	In-progress
WBS 2.2	UML Class, Sequence, Package & Activity Diagrams	Obj 1, 4	Sijian Wang, Yusai Xi	50106045, 50105989	2026-10-31	In-progress
WBS 3.1	Qwen-Turbo LLM Diagnostic Question Parser & Regex	Obj 3	Siqin Dong, Hao Jiang	50106060, 50106065	2026-11-15	In-progress
WBS 3.2	GraphRAG Retrieval Pipeline & Socratic Dialogue Core	Obj 1, 2	Hao Jiang, Yuxuan Wu	50106065, 50106070	2026-11-20	In-progress
WBS 3.3	Dynamic Quiz Generation & ECharts 5.5.1 Mastery SPA	Obj 1, 3	Yusai Xi, Siqin Dong	50105989, 50106060	2026-11-25	Pending
WBS 3.4	Pytest Test Suite, Cypher Mocks & Stress Backoff	Obj 4	Mingjie Yang, Weixin Xie	50106061, 50106034	2026-11-30	Pending
WBS 4.1	Empirical Benchmark Testing & Noise Filtering Gate	Obj 3, 4	Weixin Xie, Siqin Dong	50106034, 50106060	2026-12-05	Pending
WBS 4.2	Technical Report Authoring (40–60 pages) & Workload Profile	Obj 5	Weixin Xie, Yongtong Lin, All Members	50106034, 50106038	2026-12-08	In-progress
WBS 4.3	User Manual Authoring (8–12 pages) & PoC Deployment ZIP	Obj 1, 5	Mingjie Yang, Zixuan Liang	50106061, 50106037	2026-12-10	Pending
WBS 4.4	15-Minute MP4 Video Production & Slide Deck Preparation	Obj 5	Zixuan Liang, Yuxuan Wu, All Members	50106037, 50106070	2026-12-12	Pending
WBS 4.5	Final Deliverable Quality Gate & MyAberdeen Submission	Obj 1, 5	Yuxuan Wu	50106070	2026-12-14	Pending

References

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- [4] B. S. Bloom, "The 2 sigma problem: Methods of group instruction as effective as one-to-one tutoring," *Educ. Researcher*, vol. 13, no. 6, pp. 4–16, 1984.

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- [8] I. Sommerville, *Software Engineering*, 10th ed. Boston, MA: Pearson, 2016.